

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Practical – VIII

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8.API Creation and Integration

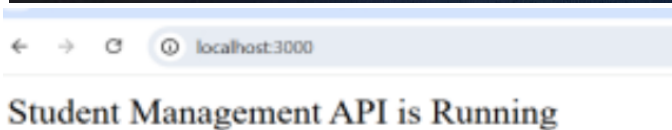
a. Create a simple REST API using Express

Code:-

```
1  const express = require("express");
2  const mongoose = require("mongoose");
3  const cors = require("cors");
4  const app = express();
5  const PORT = 3000;
6  // Middleware
7  app.use(cors());
8  app.use(express.json());
9  // Connect MongoDB
10 mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB") .then(() => {
11   console.log("MongoDB Connected Successfully"); })
12   .catch((error) => {
13     console.log("MongoDB Connection Failed");
14     console.log(error);
15   });
16 // Simple REST API
17 app.get("/", (req, res) => {
18   res.send("Student Management API is Running"); });
19 app.listen(PORT, () => {
20   console.log(`Server is running on http://localhost:${PORT}`); });
21
```

Output:-

```
PS C:\Users\vijaylaxmi\Downloads\Practical8> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
```



The screenshot shows a web browser window with the address bar set to 'localhost:3000'. The main content area of the browser displays the text 'Student Management API is Running'.

b. Implement a GET API to retrieve data.

Code:-

```
1 app.get("/students", async (req, res) => {
2   try {
3     const students = await mongoose.connection.db
4       .collection("students")
5       .find()
6       .toArray();
7     res.json(students);
8   }
9   catch (error) {
10    res.status(500).json({
11      message: "Error fetching students"
12    });
13  }
14 });
15
```

Output:-



```
[
  {
    "_id": "6aa8abb63f987f654dabaf5a",
    "rollNo": 102,
    "name": "Vijaylaxmi",
    "course": "Information Technology",
    "marks": 90,
    "grade": "A"
  },
  {
    "_id": "6aa8ac823f987f654dabaf5b",
    "rollNo": 102,
    "name": "Vijaylaxmi",
    "course": "Information Technology",
    "marks": 92,
    "grade": "A"
  },
  {
    "_id": "6aa8ac823f987f654dabaf5c",
    "rollNo": 103,
    "name": "William Moka",
    "course": "B.Sc. CS",
    "marks": 78
  }
]
```

c. Fetch data from the API in a React application.

Code:-

```
import { useEffect, useState } from "react";
import axios from "axios";

function App() {
  const [students, setStudents] = useState([]);
  useEffect(() => { axios.get("http://localhost:3000/students")
    .then((response) => {setStudents(response.data);})
```

```

.catch((error) => {console.log(error)}); }, []);
return (
  <div>
    <h1>Student List</h1>
    <table border="1">
      <thead>
        <tr>
          <th>Roll No</th>
          <th>Name</th>
          <th>Course</th>
          <th>Marks</th>
        </tr>
      </thead>
      <tbody>
        {students.map((student) => (
          <tr key={student._id}>
            <td>{student.rollNo}</td>
            <td>{student.name}</td>
            <td>{student.course}</td>
            <td>{student.marks}</td>
          </tr>
        ))}
      </tbody>
    </table>
  </div>
);
}

export default App;

```

Output:-

The screenshot shows the VS Code interface with the Terminal panel active. The left pane shows the command prompt at PS C:\Users\vijaylaxmi\Downloads\Practical18\my-react> npm run dev. The right pane shows the command prompt at PS C:\Users\vijaylaxmi\Downloads\Practical18\my-react> cd .. followed by PS C:\Users\vijaylaxmi\Downloads\Practical18> node server.js, which outputs Server is running on http://localhost:3000 and MongoDB Connected Successfully.

```
PS C:\Users\vijaylaxmi\Downloads\Practical18\my-react> npm run dev
```

```
PS C:\Users\vijaylaxmi\Downloads\Practical18\my-react> cd ..
```

```
PS C:\Users\vijaylaxmi\Downloads\Practical18> node server.js
```

```
Server is running on http://localhost:3000
```

```
MongoDB Connected Successfully
```

Student List

Roll No	Name	Course	Marks
102	Vijaylaxmi	Information Technology	90
102	Vijaylaxmi	Information Technology	92
103	William Moka	B.Sc. CS	78

d. Send data from React to the server using a POST request.

Code:-

1) Replace App.jsx code

```
import { useEffect, useState } from "react";
import axios from "axios";

function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");

  // Get students
  const getStudents = () => {
    axios.get("http://localhost:3000/students")
      .then((response) => {
        setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  };

  useEffect(() => {
    getStudents();
  }, []);

  // Add student
  const addStudent = () => {
    const newStudent = {
      rollNo: Number(rollNo),
      name: name,
      course: course,
      marks: Number(marks)
    };
    axios.post("http://localhost:3000/students", newStudent)
      .then((response) => {
        alert(response.data.message);
        setRollNo("");
        setName("");
        setCourse("");
        setMarks("");
        getStudents();
      })
  };
}
```

```
.catch((error) => {
  console.log(error);
});
};
return (
  <div>
    <h1>Student Management System</h1>
    <h2>Add Student</h2>
    <input
      type="number"
      placeholder="Roll No"
      value={rollNo}
      onChange={(e) => setRollNo(e.target.value)} />
    <br /><br />
    <input
      type="text"
      placeholder="Student Name"
      value={name}
      onChange={(e) => setName(e.target.value)} />
    <br /><br />
    <input
      type="text"
      placeholder="Course"
      value={course}
      onChange={(e) => setCourse(e.target.value)} />
    <br /><br />
    <input
      type="number"
      placeholder="Marks"
      value={marks}
      onChange={(e) => setMarks(e.target.value)} />
    <br /><br />
    <button onClick={addStudent}>
      Add Student
    </button>
    <br />
    <h2>Student List</h2>
    <table border="1">
      <thead>
        <tr>
```

```

<th>Roll No</th>
<th>Name</th>
<th>Course</th>
<th>Marks</th>
</tr>
</thead>
<tbody>
{students.map((student) => (
<tr key={student._id}>
<td>{student.rollNo}</td> <td>{student.name}</td>
<td>{student.course}</td> <td>{student.marks}</td>
</tr>
))}
</tbody>
</table>
</div>
);
}
export default App;

```

2) Add post method in server.js file above app.listen()

```

server.js > ...
app.post("/students", async (req, res) => {
  try {
    const student = req.body;
    await mongoose.connection.db
      .collection("students")
      .insertOne(student);
    res.json({
      message: "Student added successfully",
      student: student
    });
  }
  catch (error) {
    res.status(500).json({
      message: "Error adding student"
    });
  }
});

```

Output:-

← localhost:5173

Student Management System

Add Student

Roll No

Student Name

Course

Marks

Student List

Roll No	Name	Course	Marks
102	Vijaylaxmi	Information Technology	90
102	Vijaylaxmi	Information Technology	92
103	William Moka	B.Sc. CS	78
104	Soham	Biotechnology	95